

Computer Networks — Course Syllabus

Unit 1: Network Fundamentals

A computer network is a collection of interconnected devices that communicate to share resources and information. The OSI model is a conceptual framework dividing network communication into seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application, each handling a specific aspect of communication.

Unit 2: The Transport Layer

The Transport layer is responsible for end-to-end communication between hosts. TCP (Transmission Control Protocol) provides reliable, ordered, connection-oriented delivery of data using acknowledgments and retransmission. UDP (User Datagram Protocol) provides connectionless, unreliable delivery with lower overhead, suitable for applications like video streaming where speed matters more than guaranteed delivery.

Unit 3: Routing

Routing is the process of selecting a path for traffic across a network. Distance vector routing protocols, such as RIP, determine the best path based on hop count, while link state routing protocols, such as OSPF, build a complete map of the network topology to compute shortest paths using algorithms like Dijkstra's algorithm.

Unit 4: Network Security

Network security involves protecting data during transmission through techniques such as encryption, firewalls, and intrusion detection systems. Public key cryptography uses a pair of keys, public and private, to enable secure communication and digital signatures without requiring a shared secret key in advance.